

Mathematical Methods in Uncertainty Quantification (MMUQ) – L04

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MMUQ L04 – 3rd Assignment and Global Sensitivity Analysis

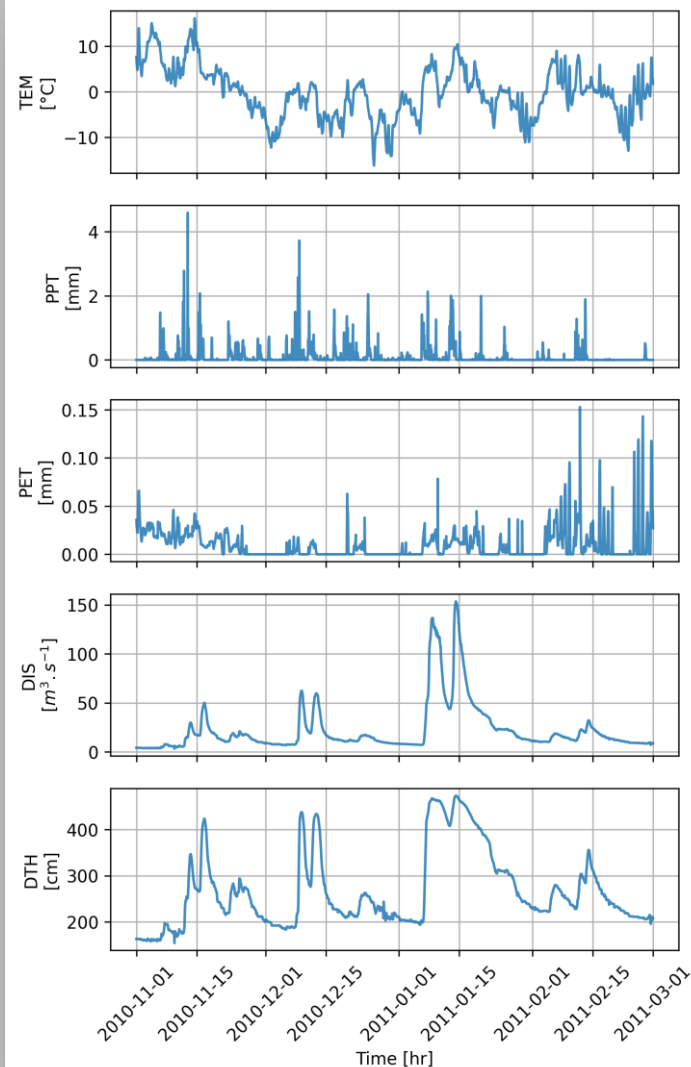
- Assignment
- Tasks
- Global Sensitivity Analysis
 - A variance-based method

Assignment – 3

- Global sensitivity analysis (GSA) of model parameters
- Present sensitivities over full range
- Present sensitivities over $\pm 20\%$ of global optimum from ASGT-1
- Reference: Uncertainty quantification, Smith 2014, Book
- Reference: Variance based sensitivity analysis of model output. Design and estimator for the total sensitivity index, Saltelli et al. 2010, Paper

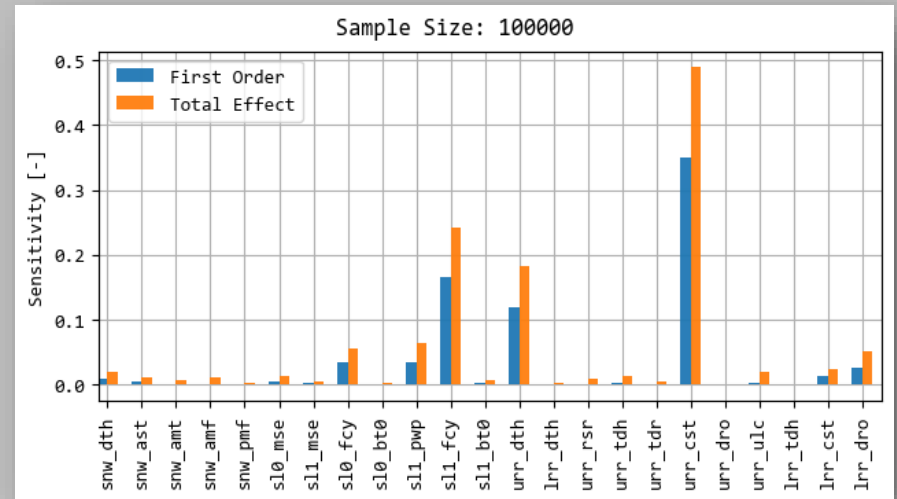
Assignment – 3, Problem

- Given
 - Data from ASGT-1
 - Optimized Xs
- Two cases
 - Full range (ASGT-1)
 - Narrow range (ASGT-2)
- Find most sensitive parameters
- Effect of changing ranges
- Effect of using a different OF
 - Compare NSE and LnNSE



Assignment – 3, Tasks

- Implement Sobol Indices
 - Smith 2014
 - Saltelli et al. 2010
- NSE and LnNSE as OFs
 - Individually
- Compute first order and total effect
 - Full range
 - Narrow range
- Compare results
- Do sensitive parameter stay the same?



Parameter Sensitivity

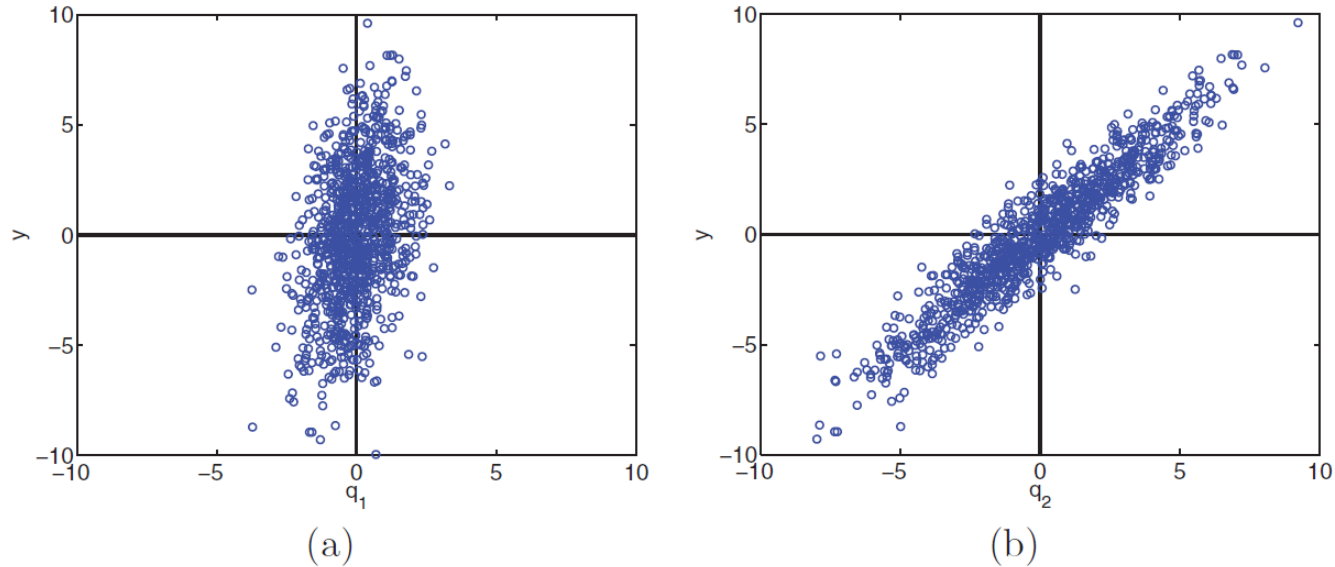


Figure 15.1. Scatterplots of y versus (a) q_1 and (b) q_2 constructed using 1000 joint realizations.

Sobol Decomposition

- Uniform IID Parameters
 - But X s have correlation!
 - Large sample size ($> 1E4$)
- Two types of indices
 - First order (S_i)
 - Total effects (S_{T_i})
- Mean (f_0)
 - Mean of OFVs
- Total Variance ($var(Y)$)
 - Variance of OFVs
- Saltelli et al. 2010
 - Better approximations

Algorithm 15.10. 1. Create two $M \times p$ sample matrices

$$A = \begin{bmatrix} q_1^1 & \cdots & q_i^1 & \cdots & q_p^1 \\ \vdots & & \vdots & & \vdots \\ q_1^M & \cdots & q_i^M & \cdots & q_p^M \end{bmatrix}, \quad B = \begin{bmatrix} \hat{q}_1^1 & \cdots & \hat{q}_i^1 & \cdots & \hat{q}_p^1 \\ \vdots & & \vdots & & \vdots \\ \hat{q}_1^M & \cdots & \hat{q}_i^M & \cdots & \hat{q}_p^M \end{bmatrix},$$

where q_i^j and $\hat{q}_i^j$ are quasi-random numbers drawn from the respective densities.

2. Create $M \times p$ matrices

$$C_i = \begin{bmatrix} \hat{q}_1^1 & \cdots & q_i^1 & \cdots & \hat{q}_p^1 \\ \vdots & & \vdots & & \vdots \\ \hat{q}_1^M & \cdots & q_i^M & \cdots & \hat{q}_p^M \end{bmatrix},$$

which are identical to B with the exception that the i^{th} column is taken from A .

3. Compute $M \times 1$ vectors of model outputs

$$y_A = f(A), \quad y_B = f(B), \quad y_{C_i} = f(C_i)$$

by evaluating the model at the input values in A, B , and C_i . The evaluation of y_A and y_B requires $2M$ model evaluations, whereas the evaluation of y_{C_i} , $i = 1, \dots, p$, requires pM evaluations. Hence the total number of model evaluations is $M(p+2)$.

4. The estimates for the first-order sensitivity indices are

$$S_i = \frac{\text{var}[\mathbb{E}(Y|q_i)]}{\text{var}(Y)} = \frac{\frac{1}{M} y_A^T y_{C_i} - f_0^2}{\frac{1}{M} y_A^T y_A - f_0^2} = \frac{\frac{1}{M} \sum_{j=1}^M y_A^j y_{C_i}^j - f_0^2}{\frac{1}{M} \sum_{j=1}^M (y_A^j)^2 - f_0^2}, \quad (15.24)$$

where the mean is approximated by

$$f_0^2 = \left(\frac{1}{M} \sum_{j=1}^M y_A^j \right) \left(\frac{1}{M} \sum_{j=1}^M y_B^j \right). \quad (15.25)$$

The estimates for the total effects indices are

$$S_{T_i} = 1 - \frac{\text{var}[\mathbb{E}(Y|q_{\sim i})]}{\text{var}(Y)} = 1 - \frac{\frac{1}{M} y_B^T y_{C_i} - f_0^2}{\frac{1}{M} y_A^T y_A - f_0^2} = 1 - \frac{\frac{1}{M} \sum_{j=1}^M y_B^j y_{C_i}^j - f_0^2}{\frac{1}{M} \sum_{j=1}^M (y_A^j)^2 - f_0^2}. \quad (15.26)$$

Sobol Decomposition

- First order (S_i)
 - Effect of X , alone, on $var(Y)$
 - Ideally, $\sum S_i = 1$
- Total effects (S_{T_i})
 - Effect of X , combined, on $var(Y)$
 - Ideally, $\sum S_{T_i} \geq 1$
 - Ideally, $S_{T_i} \geq S_i$
- The larger, the more sensitive!

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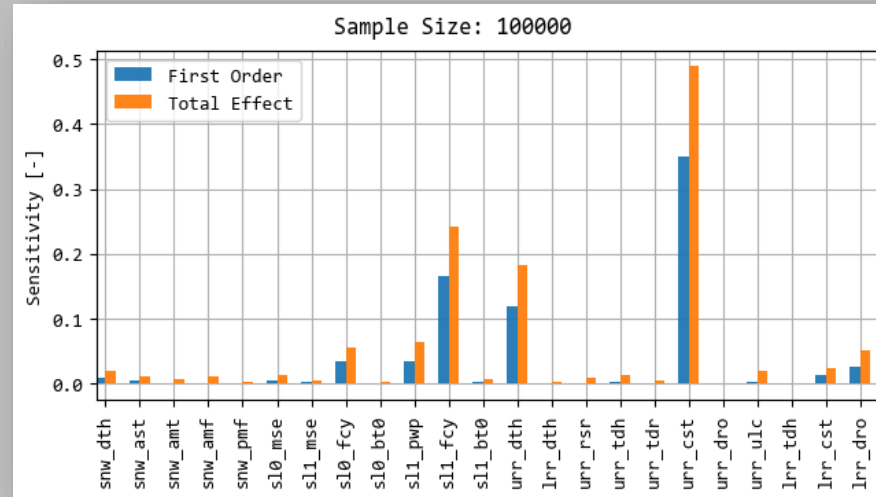
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- The larger, the more sensitive!
- $C_i = B_A^{(i)}$

Table 2

Formulas to compute S_i and S_{T_i} (in square brackets the corresponding most representative reference). Formula (b) for S_i uses the triplet $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{A}_B^{(i)}$ instead of formulation in (16), which uses $\mathbf{B}$, $\mathbf{A}$ and $\mathbf{B}_A^{(i)}$. Analogously we propose here to compute numerical estimates of (19) using quasi-random numbers in the setting $\mathbf{A}$, $\mathbf{A}_B^{(i)}$. This leads to (f) which is the *best practice* so far for S_{T_i} .

$V_{X_i}(E_{\mathbf{X}_{\sim i}}(Y X_i))$ for S_i	Reference
(a) $\frac{1}{N} \sum_{j=1}^N f(\mathbf{A})_j f(\mathbf{B}_A^{(i)})_j - f_0^2$	‘Sobol’ 1993’ [37]
(b) $\frac{1}{N} \sum_{j=1}^N f(\mathbf{B})_j (f(\mathbf{A}_B^{(i)})_j - f(\mathbf{A})_j)$	[this paper]
(c) $V(Y) - \frac{1}{2N} \sum_{j=1}^N (f(\mathbf{B})_j - f(\mathbf{A}_B^{(i)})_j)^2$	‘Jansen 1999’ [14]
$E_{\mathbf{X}_{\sim i}}(V_{X_i}(Y \mathbf{X}_{\sim i}))$ for S_{T_i}	
(d) $V(Y) - \frac{1}{N} \sum_{j=1}^N f(\mathbf{A})_j f(\mathbf{A}_B^{(i)})_j + f_0^2$	‘Homma 1996’ [11]
(e) $\frac{1}{N} \sum_{j=1}^N f(\mathbf{A})_j (f(\mathbf{A})_j - f(\mathbf{A}_B^{(i)})_j)$	‘Sobol’ 2007’ [39]
(f) $\frac{1}{2N} \sum_{j=1}^N (f(\mathbf{A})_j - f(\mathbf{A}_B^{(i)})_j)^2$	‘Jansen 1999’ [14] and [this paper]



That was it for today, thanks!